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标题	Rhodium Indenyl NHC and Fluorenyl‐Tethered NHC Half‐Sandwich Complexes: Synthesis, Structures and Applications in the Catalytic C − H Borylation of Arenes and Alkanes
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摘要（原文节选）

AbstractIndenyl (Ind) rhodium N‐heterocyclic carbene (NHC) complexes [Rh(η 5‐Ind)(NHC)(L)] were synthesised for 1,3‐bis(2,6‐diisopropylphenyl)‐4,5‐dihydroimidazol‐2‐ylidene (SIPr) with L=C2H4 (1), CO (2 a) and cyclooctene (COE; 3), for 1,3‐bis(2,4,6‐trimethylphenyl)‐4,5‐dihydroimidazol‐2‐ylidene (SIMes) with L=CO (2 b) and COE (4), and 1,3‐bis(2,4,6‐trimethylphenyl)imidazol‐2‐ylidene (IMes) with L=CO (2 c) and COE (5). Reaction of SIPr with [Rh(Cp*)(C2H4)2] did not give the desired SIPr complex, thus demonstrating ⋯

数据表 · Table 1（14 行，含数值行 14 行）

Catalyst (5 mol %) screening for the borylation of benzene with B2pin2.

Complex	NHC	L	T [° C]	t [h]	Yield [%][a]
3	SIPr	COE	80	24	97
1	SIPr	C2H4	80	24	93
2	SIPr	CO	80	48	0
[Rh(Ind)(COE)2]	–	COE	80	40	90
[RhCp*(C2H4)2]	–	C2H4	80	24	11
[RhCp*(C2H4)2]	–	C2H4	80	44	37
4	SIMes	COE	80	24	18
4	SIMes	COE	80	48	91
5	IMes	COE	80	24	21
5	IMes	COE	80	48	90
11	R=Dipp	C2H4	80	100	32
13	R=Dipp	CO	80	100	0
14	R=Mes	COE	75	100	14
16	R=Me unsaturated	COE	75	168	87

脚注（逐字）：[a] Yield determined by 1H NMR spectroscopy through integration of the product resonance against ferrocene as an internal standard. Benzene was in excess and B2pin2 was the limiting reagent.

正文对应句（逐字）：The above complexes were tested as catalysts for the C − H borylation of benzene using B2pin2 [67] with benzene in excess as the solvent (Scheme 5, Table 1). This was a good bench‐marking reaction and allowed us to ascertain that the indenyl/NHC and fluorenyl/NHC ligand sets were indeed capable of sup

数据表 · Table 2（10 行，含数值行 7 行）

Arene and alkane borylation using [Rh(Ind)(SIPr)(COE)].

Arene	T [° C], t [h], mol % of catalyst	Product	Yield [%][a]	Isomer distribution[b] (o:m:p)
benzene	80, 48, 2.5		81	–
naphthalene	80, 48, 2.5		38[c]	0.07 (1‐Bpin) : 1 (2‐Bpin)
toluene	110, 48, 2.5		76[d]	0.07 : 1.00 : 0.42
mesitylene	150, 72, 2.5		34	–
biphenyl	110, 72, 2.5		41	0.00 : 1.00 : 0.63
anisole	110, 72, 2.5		53	0.06 : 1.00 : 0.35
diphenyl ether	110, 72, 2.5		74	0.00 : 1.00 : 0.26
fluorobenzene	80, 48, 2.5		33[e]	0.91 : 1 : 0.18
n‐octane	130, 48, 5		7	terminal only
n‐decane	150, 48, 10		18	terminal only

脚注（逐字）：[a] All yields are isolated yields after column chromatography. The arene or alkane was in excess and was used as the solvent. [b] Determined by 1H NMR spectroscopy. [c] 79 % yield by 1H NMR spectroscopy. [d] 95 % yield by 1H NMR spectroscopy. [e] 95 % yield was observed by NMR spectroscopy, but thi

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